MUHAMMAD ASGHAR

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RESEARCH PROFILE

BS Chemistry graduate (Analytical Chemistry specialization) with hands-on undergraduate laboratory training in HPLC, TLC, AAS, and UV-Vis spectrophotometry, alongside wet-chemical sample preparation, solvent extraction, and quantitative data analysis. Undergraduate thesis work involved a literature-based review and analytical evaluation of chromatographic and spectroscopic methods, complemented by practical instrumental laboratory training. Recent industrial QC experience in a cement manufacturing setting. Seeking a funded, research-based Master's position in analytical, bioanalytical, or environmental chemistry, with particular interest in instrumental analysis, separation science, spectroscopy, and chemical sensors.

EDUCATION

BS in Chemistry – Analytical Chemistry Specialization Sept 2022 – May 2026
University of Education Lahore, Dera Ghazi Khan Campus, Pakistan
Medium of Instruction: English

- **Core Coursework: Advanced Analytical Instrumentation**, **HPLC & Separation Techniques**, **Instrumental Spectroscopy**, **Quantitative Chemical Analysis**, **Biochemistry**, **Coordination Chemistry**, **Organic Synthesis**

UNDERGRADUATE RESEARCH EXPERIENCE

Final-Year Research Project Jan 2026 – May 2026
Department of Chemistry, University of Education Lahore
Thesis: "A Comprehensive Review and Analytical Evaluation of Chromatographic and Spectroscopic Methods for Heavy Metals and Bioactive Compounds in Local Medicinal Plant Extracts"

- **Part I – Literature Synthesis & Method Evaluation:** Reviewed and compared chromatographic (HPLC, TLC) and spectroscopic (AAS, UV-Vis) methods reported in the literature for heavy-metal and bioactive-compound detection, evaluating sensitivity, selectivity, and applicability; assessed reported LOD/LOQ values and linear regression parameters across the surveyed studies.
- **Part II – Applied Laboratory Training:** Carried out hands-on instrumental laboratory work in parallel with the review, including HPLC and TLC method setup (mobile phase preparation, column handling, peak integration), AAS and UV-Vis operation for elemental and molecular determination, acid-digestion sample preparation, solvent extraction, and construction of multi-point calibration curves.
- Produced a full written thesis, demonstrating scientific writing, critical literature evaluation, and data visualization.

INDUSTRIAL EXPERIENCE

Quality Control Laboratory Intern June 2026 – Present
DG Khan Cement Factory, Quality Control Division, Pakistan

- Perform wet-chemical assays, gravimetric analyses, and acid-base titrations for raw-material validation under industrial compliance standards.
- Track analytical testing loops and routine instrument calibration as part of daily quality-assurance workflow.
- Maintain laboratory safety compliance, hazardous-chemical handling procedures, and analytical documentation.

TECHNICAL & LABORATORY SKILLS

- **Chromatography: HPLC** (method setup, mobile phase preparation, column handling, peak integration), **TLC**, **Column Chromatography**
- **Spectroscopy & Elemental Analysis: Atomic Absorption Spectroscopy (AAS)**, **UV-Vis Spectrophotometry**, **Flame Photometry**; working familiarity with **ICP-OES** principles
- **Wet Chemistry & Sample Preparation: Acid Digestion**, **Liquid-Liquid Extraction (LLE)**, **Solid-Phase Extraction (SPE)**, **Titrimetric Analysis** (acid-base, redox, complexometric)
- **Data & Software:** OriginPro, ChemDraw, Microsoft Excel, LaTeX, EndNote, Mendeley

CONFERENCE PRESENTATION

1st International Conference on Innovations in Chemistry, Biochemistry, Biotechnology and Bioinformatics (ICBBB-2022) Dec 2022
The Islamia University of Bahawalpur, Pakistan

- **Poster:** "DFT-Based Modeling of Asymmetric Non-Fullerene Acceptors for High-Performance Organic Solar Cells." An early undergraduate project in computational structure-property analysis, reflecting a broader interest in structure-based analytical reasoning alongside the experimental analytical chemistry focus developed later in the degree.

CERTIFICATIONS & LANGUAGE PROFICIENCY

- **U.S. Department of State English Access Microscholarship Program** (2024–2026), U.S. Embassy Islamabad – competitive, fully funded 2-year merit program in advanced academic writing, research reporting, and presentation skills.
- **English Proficiency (self-assessed, CEFR scale):** Listening C2, Spoken Interaction C2, Spoken Production C2, Reading C1, Writing C1
- **Native / Additional Languages:** Urdu, Saraiki, Punjabi

ACADEMIC REFERENCES

Dr. Rana Farhat Mehmood
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Dr. Imran Touseef
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